

networks of a different family of IP addresses. A network with a router to bridge two families of networks is shown in Figure 2.

We will now build a router using a virtual machine. First, start two virtual machines with network interfaces of different collision domains. Then set the IP addresses of the different families. Start a virtual machine with two interfaces of collision domains, and set IP addresses for the interfaces.

```
slynux@gnubox:~$ vstart pc1 --eth0=A // PC1
slynux@gnubox:~$ vstart pc1 --eth0=B // PC2
slynux@gnubox:~$ vstart router --eth0=A --eth1=B // Router
```

Execute the following commands in the virtual machines.

On R1:

```
router# ifconfig eth0 192.168.0.1
router# ifconfig eth1 192.168.1.1
```

On PC1:

```
pc1# ifconfig eth0 192.168.0.2
pc1# route add default gw 192.168.0.1 // Setting default gateway
```

On PC2:

```
pc1# ifconfig eth0 192.168.1.2
pc1# route add default gw 192.168.1.1 // Setting default gateway
```

Run the following ping tests to check if things are in place:

```
pc1# ping 192.168.0.1
pc1# ping 192.168.1.1
pc1# ping 192.168.1.2
```

```
pc2# ping 192.168.1.1
pc2# ping 192.168.0.1
pc2# ping 192.168.0.2
```

Now execute the `tracert` command to find out the path of the network packet transfer.

On PC1:

```
pc1:~$ tracert 192.168.1.2
tracert to 192.168.1.2 (192.168.1.2), 64 hops max, 40 byte packets
 1 192.168.0.1 (192.168.0.1) 4 ms 3 ms 6 ms
 2 192.168.1.5 (192.168.1.2) 1 ms 1 ms 1 ms
```

This means that it first passes through the gateway 192.168.0.1 and reaches the destination.

Figure 3 shows a network with two routers for

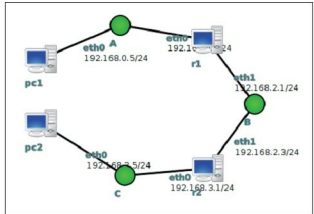


Figure 3: A network with two routers for advanced routing of different addresses

advanced routing of different addresses.

This experiment is more interesting, since we will use two routers and a bunch of `route` command executions to specify more address routes.

Let's start two VMs with interfaces belonging to collision domains A and C, and two routers R1 and R2 with collision domains of the interfaces A, B and B, C.

Execute the following commands:

```
slynux@gnubox:~$ vstart pc1 --eth0=A
slynux@gnubox:~$ vstart pc2 --eth0=C
slynux@gnubox:~$ vstart r1 --eth0=A --eth1=B
slynux@gnubox:~$ vstart r2 --eth0=B --eth1=C
```

On PC1:

```
pc1# ifconfig eth0 192.168.0.5
pc2# ifconfig eth0 192.168.3.5
```

```
r1# ifconfig eth0 192.168.0.1
r1# ifconfig eth1 192.168.2.1
```

```
r2# ifconfig eth1 192.168.2.3
r2# ifconfig eth0 192.168.3.1
```

Add the `routes`:

```
r1# route add -net 192.168.3.0 netmask 255.255.255.0 gw 192.168.2.1 dev eth0
r2# route add -net 192.168.0.0 netmask 255.255.255.0 gw 192.168.2.3 dev eth1
```

```
pc1# ping 192.168.3.5
pc2# ping 192.168.0.1
```

Let's now connect a network with the host machine gateway and access the Internet.

Each of the virtual machines used here are Debian Sid-based GNU/Linux installations. So we can update and install custom tools by providing connectivity